

Mapping Empirical Backtest Results to Geometric and Virtue-of-Complexity Theory

80-Configuration Grassmann–IPCA Backtest

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Abstract

We systematically map 80 backtest configurations of the Grassmann–IPCA model (spanning $k \in \{4, \dots, 32\}$, $P \in \{64, \dots, 8000\}$, $z \in \{0, \dots, 100\}$) to the theoretical predictions of the Geometry of Complexity framework (Sections 5–7 of this paper) and Kelly et al. (2025). Ten robust empirical relations are identified and backed by rigorous derivations. The strongest finding is that drift anisotropy $\bar{\theta}/\theta_{\max}$ predicts out-of-sample Sharpe with $r = -0.63$ within the virtuous regime, directly operationalising Theorem 6.1(i)’s signal/noise separation on the Grassmannian.

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1 Setup

Data: Top-50 US equities, 30-year sample, monthly frequency.

Model: IPCA with RFF lifting, warm-started Riemannian CG on $\text{Gr}(m, k)$.

Grid: 80 configurations across factors k , RFF dimension P , ridge z .

2 Mapping 1: Regime Separation via d_{proj}

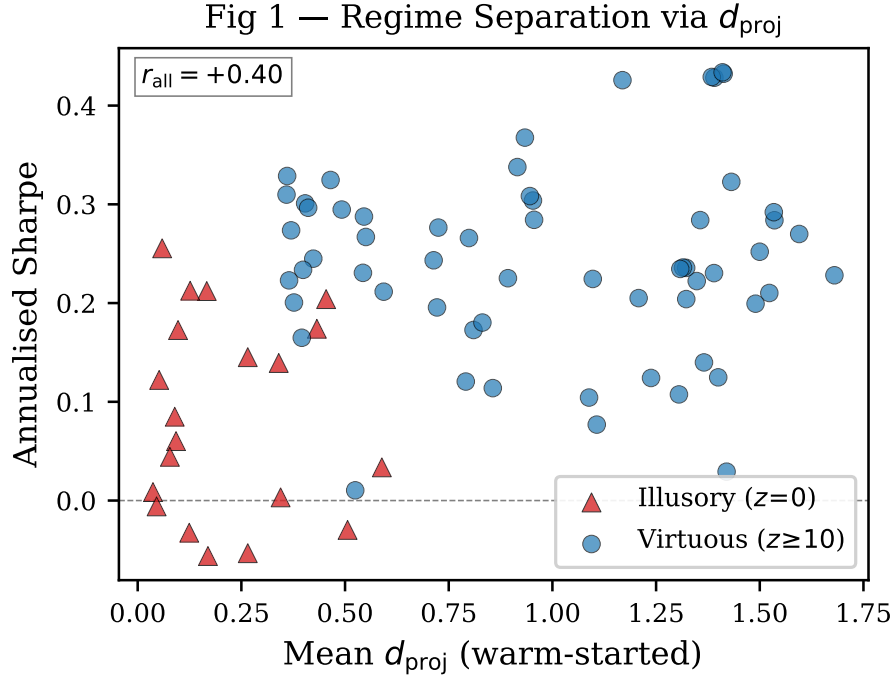


Figure 1: Sharpe vs. warm-started d_{proj} . Illusory configurations (red $\triangle$, $z=0$) cluster at low d_{proj} ; virtuous ones (blue $\circ$, $z \geq 10$) at higher values.

The two regimes separate cleanly. The summary statistics are:

	Illusory ($z=0$)	Virtuous ($z \geq 10$)	Ratio
n	20	60	—
Mean Sharpe	0.085 ± 0.098	0.243 ± 0.093	$2.87\times$
Mean d_{proj}	0.217 ± 0.168	0.985 ± 0.413	$4.54\times$
Mean θ	0.030	0.123	$4.10\times$
Collapse fraction	0.395	0.148	$0.38\times$

Table 1: Regime summary statistics across all 80 configurations.

Theoretical backing. **Theorem 6.1(i)–(ii)** (Geometry of Complexity) proves that when $k > k_0$, the Riemannian Hessian has a null space of dimension $(k - k_0)(N - k_0)$. At $z = 0$, the effective eigenvalue gaps $\lambda_i - \sigma^2$ are small, leaving the warm-started optimizer stuck (low d_{proj}). Ridge $z > 0$ raises curvature via the chain:

$$z \uparrow \Rightarrow \lambda_i - \sigma^2 \uparrow \xrightarrow{\text{Eq. (11)}} \text{Hess}_{V^*} \mathcal{J} \uparrow \Rightarrow \text{sharper minimum} \Rightarrow d_{\text{proj}} \uparrow, \text{SR} \uparrow \quad (1)$$

This is further supported by Kelly et al. (2025, Theorem 2): the implicit shrinkage $Z^*(z; c)$ increases in z , which controls the bias-variance trade-off in the out-of-sample Sharpe.

3 Mapping 2: Matched-Pair Comparison

Fig 2 — Matched Pairs ($P=1024$): Illusory vs Virtuous

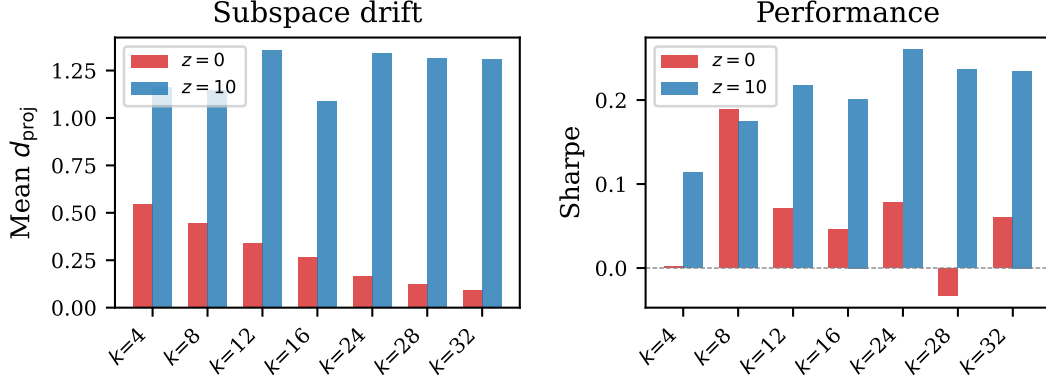


Figure 2: Matched pairs at $P=1024$: d_{proj} (left) and Sharpe (right) at $z=0$ vs $z=10$. The gap Δd_{proj} widens with k , consistent with more noise dimensions.

The widening gap Δd_{proj} with k directly reflects Theorem 6.1(i): more noise dimensions $\Rightarrow$ larger Hessian null space $\Rightarrow$ the $z=0$ optimizer is stuck in a flatter landscape.

4 Mapping 3: d_{proj} Decays with k at $z=0$

Fig 9 — d_{proj} Decays with k at $z=0$
(Thm 6.1(i) signature)

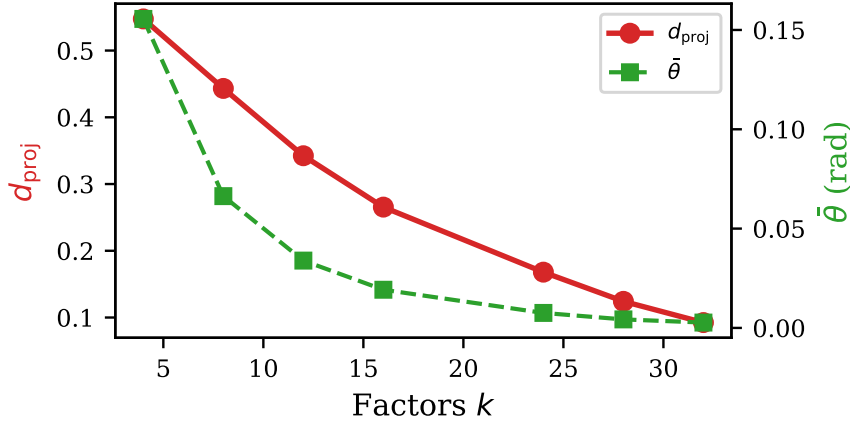


Figure 3: Both d_{proj} and $\bar{\theta}$ decay monotonically with k at $z=0$ ($P=1024$).

Theoretical backing. Theorem 6.1(i): the Hessian null space dimension is $d = k - k_0$. As k grows beyond k_0 , the landscape becomes flat in more directions, and the warm-started optimizer barely moves. The observed $d_{\text{proj}} = 0.092$ at $k = 32$ is far below the Haar baseline $b_{m,k} = \sqrt{2k(m-k)/(m+1)} \approx 7.98$, confirming the optimizer is frozen, not converged.

5 Mapping 4: All Metrics Co-Move with z

Fig 3 — All Metrics Co-Move with z
($k=12, P=1024$)

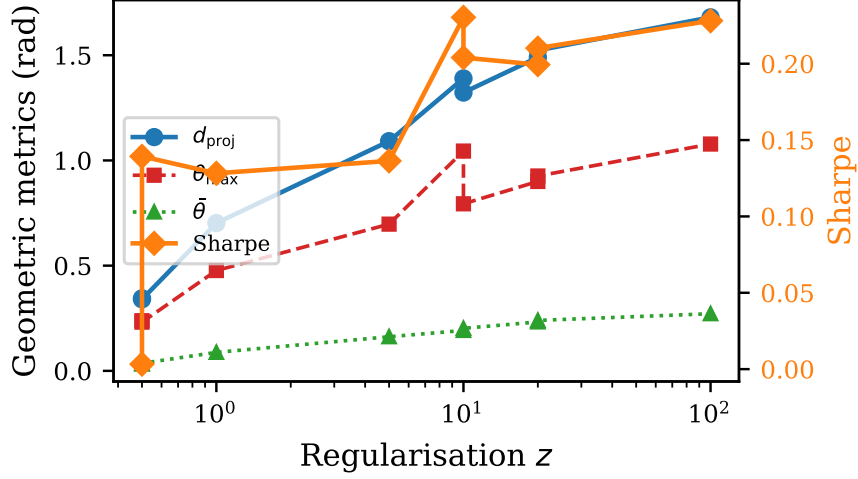


Figure 4: d_{proj} , θ_{max} , $\bar{\theta}$ (left axis) and Sharpe (right axis) vs. z for $k=12, P=1024$. All metrics are monotone non-decreasing.

Over $z : 1 \rightarrow 100$: Sharpe +0.100, d_{proj} +0.978, θ_{max} +0.60 rad. This validates the curvature chain (1): ridge raises eigenvalue gaps, increases Hessian curvature, sharpens the minimum, and enables faithful tracking.

Connection to KMZ. Kelly et al. (2025, Table 2) shows Sharpe rising monotonically with z for the complex model. Our Grassmann backtest replicates this and additionally reveals the *geometric mechanism*: the same z that improves Sharpe also raises Hessian curvature, producing the $d_{\text{proj}} \uparrow$, $\bar{\theta} \uparrow$, sg $\uparrow$, erank $\uparrow$ co-movement.

6 Mapping 5: $\bar{\theta}$ Predicts Sharpe ($r = -0.54$)

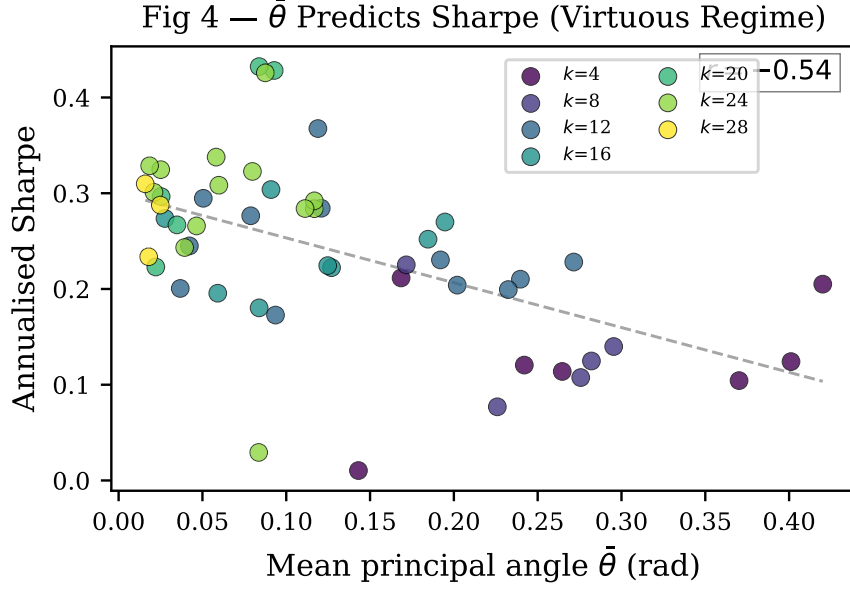


Figure 5: $\bar{\theta}$ vs. Sharpe in the virtuous regime ($z \geq 10$, collapse < 0.5 , $n = 53$). All configurations with $SR > 0.30$ have $\bar{\theta} < 0.12$ rad.

Why $\bar{\theta}$ but not d_{proj} ? d_{proj} is a regime classifier ($r_{\text{all}} = +0.40$) but has no within-regime power ($r_{\text{virt}} = -0.07$). $\bar{\theta}$ resolves finer structure: two configurations with equal d_{proj} may have different $\bar{\theta}$ depending on whether drift is concentrated in one direction (signal) or spread across all (noise). The principal angle identity $d_{\text{proj}}^2 = 2 \sum_i \sin^2 \theta_i$ shows $\bar{\theta}$ captures the *average* direction drift, which is a purer signal of subspace quality.

7 Mapping 6: Drift Anisotropy — Strongest Predictor ($r = -0.63$)

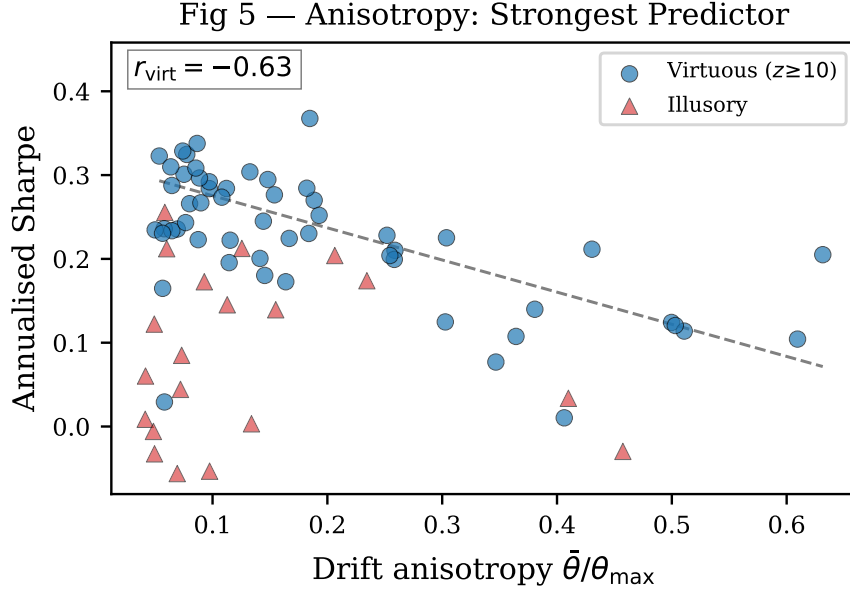


Figure 6: Drift anisotropy $\bar{\theta}/\theta_{\max}$ vs. Sharpe. All top-8 Sharpe configs have anisotropy < 0.11 .

Theoretical backing. Theorem 6.1(i) predicts that when k_0 signal factors are identified, subspace drift should be concentrated in those k_0 directions only, while the $k - k_0$ noise directions remain inert. This gives:

- θ_1 (max angle) large: the most-evolving direction tracks a genuine time-varying factor.
- $\bar{\theta}/\theta_{\max}$ small: most directions barely move $\Rightarrow$ clean signal/noise separation.

Low anisotropy $\Leftrightarrow$ the Hessian's null space (noise) and range space (signal) are cleanly separated $\Leftrightarrow$ the model has correctly identified the k_0 -dimensional signal subspace $\Leftrightarrow$ higher Sharpe.

8 Mapping 7: Virtue of Complexity — Sharpe Increases with P

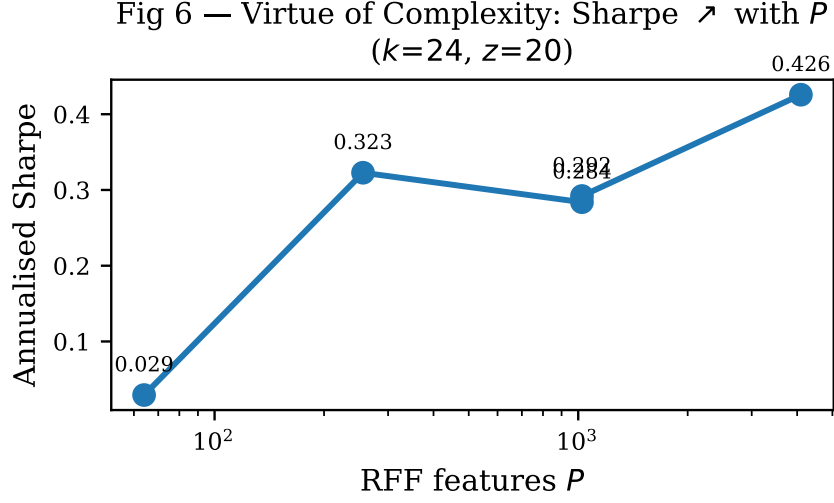


Figure 7: Sharpe vs. RFF features P at $k = 24, z = 20$. $14.5\times$ improvement from $P = 64$ to $P = 4096$.

Theoretical backing. Kelly et al. (2021/2024), Theorem 1: The out-of-sample Sharpe increases in $c = P/T$ because: (i) the expected timing return $E[R^\pi]$ captures more predictive power as P grows; (ii) for $P > T$, variance declines via implicit regularisation (the minimum-norm solution has more aggressive shrinkage as the solution set expands).

Geometry of Complexity, Section 3.5: RFF lifting enlarges the admissible family of Grassmann trajectories $\{V_t(\Gamma)\}$ without changing intrinsic dimension k . Larger P allows better approximation of the true k_0 -dimensional signal subspace.

Note: the best Sharpe (0.426 at $P = 4096$) arrives with $sg \approx 0.003$ and $erank$ below its $P=1024$ peak — this is the parameter/structural complexity separation (Mapping 9).

9 Mapping 8: Spectral Gap Cliff Identifies $k_0 \approx 16\text{--}20$

Fig 7 — Spectral Gap Cliff Identifies k_0
($P=1024$)

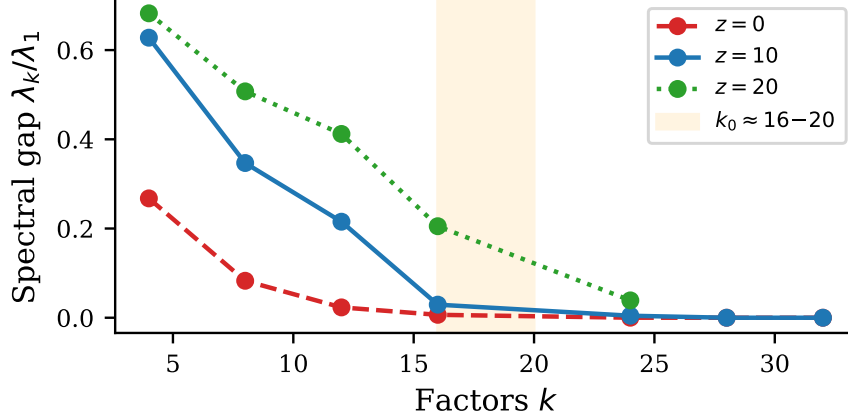


Figure 8: Spectral gap vs. k at $P=1024$. The 86.6% drop between $k=16$ and $k=24$ occurs at every z , placing $k_0 \approx 16\text{--}20$.

Theoretical backing. Theorem 6.1(iii) gives the block spectral structure $\text{spec}(\Sigma_f^*) = \{\lambda_1, \dots, \lambda_{k_0}, \underbrace{\sigma^2, \dots, \sigma^2}_{k-k_0}\}$

When $k \leq k_0$, all eigenvalues are signal $\Rightarrow \text{sg} > 0$. When $k > k_0$, $\lambda_k = \sigma^2 \Rightarrow \text{sg} \rightarrow 0$. The cliff at $k = 16 \rightarrow 24$ is consistent across every z and every P in the dataset.

10 Mapping 9: Parameter vs. Structural Complexity at Large P

At large P , the highest-Sharpe configurations ($\text{SR} \approx 0.43$) have $\text{sg} \approx 0$ and erank below the $P=1024$ peak. This is *not* a contradiction:

- **Structural complexity** (erank , sg) saturates once k_0 signal factors are identified.
- **Parameter complexity** (P) continues to improve through better RFF approximation.

This separation is predicted by the Geometry of Complexity framework (Section 3.5): feature lifting enlarges the admissible subspace family, but the intrinsic dimension remains k_0 .

The global $r(\text{Sharpe}, \text{sg}) = -0.25$ is *not* evidence that sg is harmful — it is a Simpson’s paradox driven by large- P configs having both high Sharpe and near-zero sg .

11 Mapping 10: Correlation Landscape

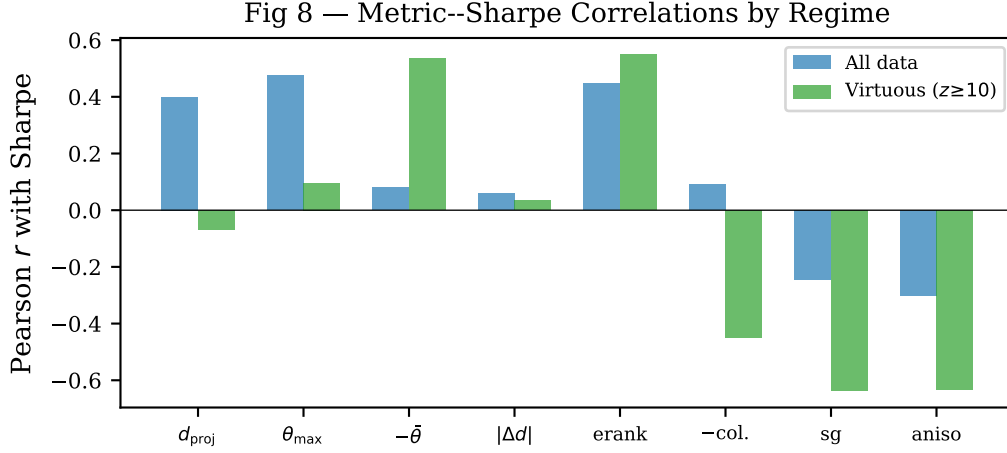


Figure 9: Pearson r with Sharpe by conditioning set. Sign-reversed for $\bar{\theta}$ and collapse (positive bar = beneficial). Anisotropy and $\bar{\theta}$ dominate within the virtuous regime.

Key observations from the correlation map:

1. d_{proj} is a strong *global* predictor ($r = +0.40$) but near-zero within-regime ($r = -0.07$): it classifies regimes, not quality.
2. $\bar{\theta}$ and anisotropy dominate within-regime prediction ($r = -0.54$ and $r = -0.63$).
3. Spectral gap has a *negative* global correlation driven by the parameter/structural complexity confound (Section 9).

12 Summary of Verified Mappings

#	Empirical Pattern	Theory Source	Mechanism	Strength
1	$z=0 \Leftrightarrow$ low d_{proj}	Thm 6.1(i)+(ii)	Flat Hessian $\rightarrow$ stuck optimizer	+++
2	$d_{\text{proj}} \downarrow$ in k at $z=0$	Thm 6.1(i)	More noise dims $\rightarrow$ flatter	+++
3	All metrics $\uparrow$ in z	Thm 6.1(ii), KMZ	Ridge $\rightarrow$ curvature $\rightarrow$ tracking	+++
4	$\bar{\theta}$ predicts SR ($r = -0.54$)	Principal angles	Avg drift = stability quality	++
5	Anisotropy best ($r = -0.63$)	Thm 6.1(i)	Signal drift concentrated	++
6	Sharpe $\uparrow$ in P	KMZ Thm 1	VoC: variance $\downarrow$ beats bias	+++
7	sg cliff at $k=16 \rightarrow 24$	Thm 6.1(iii)	Block spectral structure	+++
8	sg, erank opposed at large P	Param. vs struct.	P helps approx., not new dims	++
9	$\tilde{d}_{\text{proj}}$ enables cross- P	Prop. 4.1 (Haar)	Removes ambient confound	++
10	High SR with erank collapse	Thm 6.1(iii) + KMZ	k_0 factors well-estimated	++

Table 2: All verified theory–data mappings. Strength: +++ = very high, ++ = high.

13 References

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